

Physics 513 QFT: HW 9

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1. 1

I use the notation in my QED and Real Scalar Field notes. My notation matches Schwartz. I don't like using π for a field so I relabel $\pi \rightarrow \phi$. The interaction Lagrangian is

$$\mathcal{L}_1 = ig\bar{\psi}\gamma_5\psi\phi. \quad (1)$$

1.1. Feynman Rules

I represent the electron field ψ with a straight line: $\longrightarrow$. The electron field propagator is

$$i \frac{\not{p} + m}{p^2 - m^2 + i\epsilon}, \quad (2)$$

where m is the electron mass.

The pseudoscalar field ϕ is represented by a dashed line $--\longrightarrow$. It's propagator is the same as the scalar field propagator

$$\frac{i}{p^2 - M^2 + i\epsilon}, \quad (3)$$

where M is the ϕ boson mass.

The vertex joins two fermions with one boson (Fig. 1) and contributes the factor $-g\gamma_5$.

1.2. (1) $e^- \phi \rightarrow e^- \phi$

$$\vec{p}_1(e^-) + \vec{p}_2(\phi) \rightarrow \vec{p}_3(e^-) + \vec{p}_4(\phi) \quad (4)$$

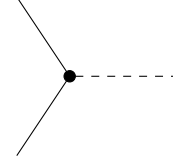


FIG. 1. Interaction vertex.

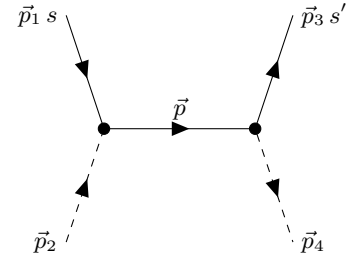


FIG. 2. ☿

Since this is a 2 to 2 reaction I use Mandelstam variables

$$\begin{aligned} s &= (\vec{p}_1 + \vec{p}_2)^2 = (\vec{p}_3 + \vec{p}_4)^2, \\ t &= (\vec{p}_1 - \vec{p}_3)^2 = (\vec{p}_2 - \vec{p}_4)^2, \\ u &= (\vec{p}_1 - \vec{p}_4)^2 = (\vec{p}_2 - \vec{p}_3)^2. \end{aligned} \quad (5)$$

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{K} = \vec{p}_3 + \vec{p}_4 \quad (6)$$

The matrix element (non-trivial part of S-matrix) is

$$i\delta^{(4)}(\vec{P} - \vec{K})\mathcal{M} = \text{☿} + \text{☿}, \quad (7)$$

where I give the Feynman diagrams in Figs. 2 & 3.

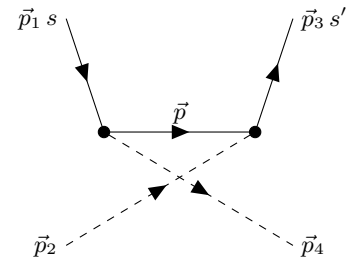


FIG. 3. Feynman diagram ☿. Vertices have dots, crossings do not.

$$\begin{aligned} \mathfrak{Y} = & ig^2 \int d^4p \bar{u}_{s'}(\vec{p}_3) \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} \gamma_5 u_s(\vec{p}_1) \\ & \delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}) \delta^{(4)}(\vec{p} - \vec{p}_3 - \vec{p}_4) \end{aligned} \quad (8)$$

$$\begin{aligned} \mathfrak{Y} = & ig^2 \int d^4p \bar{u}_{s'}(\vec{p}_3) \gamma_5 \frac{\gamma_\mu p^\mu + m}{\vec{p}^2 - m^2 + i\epsilon} \gamma_5 u_s(\vec{p}_1) \\ & \delta^{(4)}(\vec{P} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{K}) \end{aligned} \quad (9)$$

$$\mathfrak{Y} = i\delta^{(4)}(\vec{P} - \vec{K}) g^2 \bar{u}_{s'}(\vec{p}_3) \gamma_5 \frac{\gamma_\mu P^\mu + m}{\vec{P}^2 - m^2} \gamma_5 u_s(\vec{p}_1) \quad (10)$$

$$\mathfrak{Y} = i\delta^{(4)}(\vec{P} - \vec{K}) g^2 \bar{u}_{s'}(\vec{p}_3) \gamma_5 \frac{\not{P} + m}{\vec{P}^2 - m^2} \gamma_5 u_s(\vec{p}_1) \quad (11)$$

$$\mathfrak{Y} = i\delta^{(4)}(\vec{P} - \vec{K}) g^2 \bar{u}_{s'}(\vec{p}_3) \gamma_5 \frac{\not{P} + m}{s - m^2} \gamma_5 u_s(\vec{p}_1) \quad (12)$$

$$\begin{aligned} \mathfrak{Y} = & ig^2 \int d^4p \bar{u}_{s'}(\vec{p}_3) \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} \gamma_5 u_s(\vec{p}_1) \\ & \delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_3) \end{aligned} \quad (13)$$

$$\begin{aligned} \mathfrak{Y} = & ig^2 \int d^4p \bar{u}_{s'}(\vec{p}_3) \gamma_5 \frac{\gamma_\mu p^\mu + m}{\vec{p}^2 - m^2 + i\epsilon} \gamma_5 u_s(\vec{p}_1) \\ & \delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_3) \end{aligned} \quad (14)$$

$$\mathfrak{Y} = i\delta^{(4)}(\vec{P} - \vec{K}) g^2 \bar{u}_{s'}(\vec{p}_3) \gamma_5 \frac{\gamma_\mu (p_1^\mu - p_4^\mu) + m}{(\vec{p}_1 - \vec{p}_4)^2 - m^2} \gamma_5 u_s(\vec{p}_1) \quad (15)$$

$$\mathfrak{Y} = i\delta^{(4)}(\vec{P} - \vec{K}) g^2 \bar{u}_{s'}(\vec{p}_3) \gamma_5 \frac{\not{p}_1 - \not{p}_4 + m}{u - m^2} \gamma_5 u_s(\vec{p}_1) \quad (16)$$

$$\boxed{\frac{\mathcal{M}}{g^2} = \bar{u}_{s'}(\vec{p}_3) \gamma_5 \left(\frac{\not{P} + m}{s - m^2} + \frac{\not{p}_1 - \not{p}_4 + m}{u - m^2} \right) \gamma_5 u_s(\vec{p}_1)} \quad (17)$$

1.3. (2) $e^+ \phi \rightarrow e^+ \phi$

$$\vec{p}_1(e^+) + \vec{p}_2(\phi) \rightarrow \vec{p}_3(e^+) + \vec{p}_4(\phi) \quad (18)$$

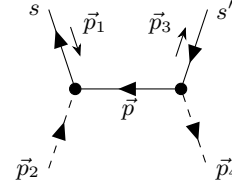


FIG. 4. $\mathfrak{Y}$. I only show momentum arrows when they contradict particle flow (i.e. only for positrons).

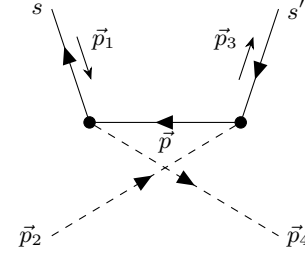


FIG. 5. $\mathfrak{Y}$

$$i\delta^{(4)}(\vec{P} - \vec{K}) \mathcal{M} = \mathfrak{Y} + \mathfrak{Y} \quad (19)$$

The diagrams are in Figs. 4 & 5.

$$\begin{aligned} \mathfrak{Y} = & ig^2 \int d^4p \bar{v}_s(\vec{p}_1) \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} \gamma_5 v_{s'}(\vec{p}_3) \\ & \delta^{(4)}(\vec{p}_1 + \vec{p}_2 + \vec{p}) \delta^{(4)}(-\vec{p} - \vec{p}_3 - \vec{p}_4) \end{aligned} \quad (20)$$

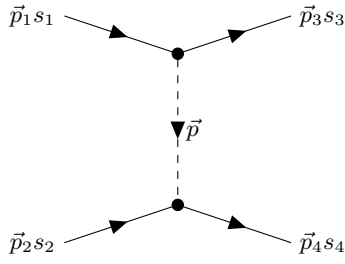
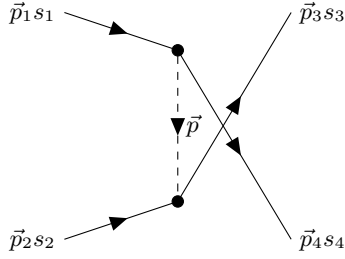
$$\begin{aligned} \mathfrak{Y} = & ig^2 \int d^4p \bar{v}_s(\vec{p}_1) \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} \gamma_5 v_{s'}(\vec{p}_3) \\ & \delta^{(4)}(\vec{P} + \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_3 + \vec{p}_4) \end{aligned} \quad (21)$$

$$\begin{aligned} \mathfrak{Y} = & ig^2 \int d^4p \bar{v}_s(\vec{p}_1) \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} \gamma_5 v_{s'}(\vec{p}_3) \\ & \delta^{(4)}(\vec{P} + \vec{p}) \delta^{(4)}(\vec{p} + \vec{K}) \end{aligned} \quad (22)$$

$$\mathfrak{Y} = ig^2 \bar{v}_s(\vec{p}_1) \gamma_5 \frac{m - \not{P}}{\vec{P}^2 - m^2} \gamma_5 v_{s'}(\vec{p}_3) \delta^{(4)}(-\vec{P} + \vec{K}) \quad (23)$$

$$\mathfrak{Y} = i\delta^{(4)}(\vec{P} - \vec{K}) g^2 \bar{v}_s(\vec{p}_1) \gamma_5 \frac{m - \not{P}}{s - m^2} \gamma_5 v_{s'}(\vec{p}_3) \quad (24)$$

$$\begin{aligned} \mathfrak{Y} = & ig^2 \int d^4p \bar{v}_s(\vec{p}_1) \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} \gamma_5 v_{s'}(\vec{p}_3) \\ & \delta^{(4)}(\vec{p} + \vec{p}_1 - \vec{p}_4) \delta^{(4)}(\vec{p}_2 - \vec{p}_3 - \vec{p}) \end{aligned} \quad (25)$$

FIG. 6. $\bar{\phi}$.FIG. 7. $\bar{\phi}$.

$$\bar{\phi} = i\delta^{(4)}(\vec{P} - \vec{K})g^2\bar{v}_s(\vec{p}_1)\gamma_5\frac{\not{p}_4 - \not{p}_1 + m}{(\vec{p}_4 - \vec{p}_1)^2 - m^2}\gamma_5 v_{s'}(\vec{p}_3) \quad (26)$$

$$\bar{\phi} = i\delta^{(4)}(\vec{P} - \vec{K})g^2\bar{v}_s(\vec{p}_1)\gamma_5\frac{\not{p}_4 - \not{p}_1 + m}{u - m^2}\gamma_5 v_{s'}(\vec{p}_3) \quad (27)$$

$$\boxed{\frac{\mathcal{M}}{g^2} = \bar{v}_s(\vec{p}_1)\gamma_5\left(\frac{m - \not{p}}{s - m^2} + \frac{\not{p}_4 - \not{p}_1 + m}{u - m^2}\right)\gamma_5 v_{s'}(\vec{p}_3)} \quad (28)$$

1.4. (3) $e^-e^- \rightarrow e^-e^-$

$$\vec{p}_1(e^-) + \vec{p}_2(e^-) \rightarrow \vec{p}_3(e^-) + \vec{p}_4(e^-) \quad (29)$$

$$i\delta^{(4)}(\vec{P} - \vec{K})\mathcal{M} = \bar{\phi} - \bar{\phi} \quad (30)$$

The diagrams are in Figs. 6 & 7. The relative minus sign comes from swapping external electrons between diagrams. Overall sign does not matter, because observables depend on $|\mathcal{M}|^2$, but relative sign does.

For this reaction I use the shorthand $u_n = u_{s_n}(\vec{p}_n)$.

$$\bar{\phi} = ig^2 \int d^4p \frac{\bar{u}_3\gamma_5 u_1 \bar{u}_4\gamma_5 u_2}{\vec{p}^2 - M^2 + i\epsilon} \delta^{(4)}(\vec{p}_1 - \vec{p}_3 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_4) \quad (31)$$

$$\bar{\phi} = i\delta^{(4)}(\vec{P} - \vec{K})g^2 \frac{\bar{u}_3\gamma_5 u_1 \bar{u}_4\gamma_5 u_2}{(\vec{p}_1 - \vec{p}_3)^2 - M^2} \quad (32)$$

$$\bar{\phi} = i\delta^{(4)}(\vec{P} - \vec{K})g^2 \frac{\bar{u}_3\gamma_5 u_1 \bar{u}_4\gamma_5 u_2}{t - M^2} \quad (33)$$

$$\bar{\phi} = ig^2 \int d^4p \frac{\bar{u}_4\gamma_5 u_1 \bar{u}_3\gamma_5 u_2}{\vec{p}^2 - M^2 + i\epsilon} \delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_3) \quad (34)$$

$$\bar{\phi} = i\delta^{(4)}(\vec{P} - \vec{K})g^2 \frac{\bar{u}_4\gamma_5 u_1 \bar{u}_3\gamma_5 u_2}{(\vec{p}_1 - \vec{p}_4)^2 - M^2} \quad (35)$$

$$\bar{\phi} = i\delta^{(4)}(\vec{P} - \vec{K})g^2 \frac{\bar{u}_4\gamma_5 u_1 \bar{u}_3\gamma_5 u_2}{u - M^2} \quad (36)$$

$$\frac{\mathcal{M}}{g^2} = \frac{\bar{u}_3\gamma_5 u_1 \bar{u}_4\gamma_5 u_2}{t - M^2} - \frac{\bar{u}_4\gamma_5 u_1 \bar{u}_3\gamma_5 u_2}{u - M^2} \quad (37)$$

$$\boxed{\frac{\mathcal{M}}{g^2} = \bar{u}_4\gamma_5 \left(\frac{u_2 \bar{u}_3\gamma_5 u_1}{t - M^2} - \frac{u_1 \bar{u}_3\gamma_5 u_2}{u - M^2} \right)} \quad (38)$$

1.5. (4) $e^-e^+ \rightarrow e^-e^+$

$$\vec{p}_1(e^-) + \vec{k}_1(e^+) \rightarrow \vec{p}_2(e^-) + \vec{k}_2(e^+) \quad (39)$$

For this reaction the Mandelstam variables are

$$\begin{aligned} s &= (\vec{p}_1 + \vec{k}_1)^2 = (\vec{p}_2 + \vec{k}_2)^2, \\ t &= (\vec{p}_1 - \vec{p}_2)^2 = (\vec{k}_1 - \vec{k}_2)^2, \\ u &= (\vec{p}_1 - \vec{k}_2)^2 = (\vec{k}_1 - \vec{p}_2)^2. \end{aligned} \quad (40)$$

The total momenta are

$$\vec{P} = \vec{p}_1 + \vec{k}_1, \quad \vec{K} = \vec{p}_2 + \vec{k}_2 \quad (41)$$

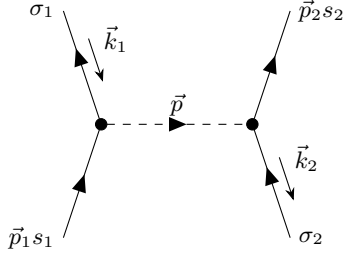
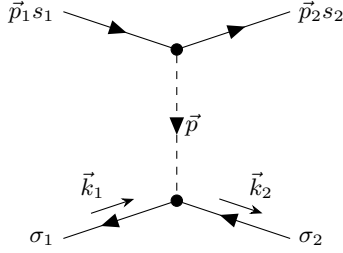
$$i\delta^{(4)}(\vec{P} - \vec{K})\mathcal{M} = \bar{\phi} - \bar{\phi} \quad (42)$$

The diagrams are in Figs. 8 & 9. The relative minus sign comes from rotating the $\bar{\phi}$ diagram, swapping an electron and positron (no sign change for that), then swapping the positrons (sign change here) to get $\bar{\phi}$. I use the shorthands $v_n = v_{\sigma_n}(\vec{k}_n)$ and $u_n = u_{s_n}(\vec{p}_n)$.

$$\bar{\phi} = ig^2 \int d^4p \frac{\bar{v}_1\gamma_5 u_1 \bar{u}_2\gamma_5 v_2}{\vec{p}^2 - M^2 + i\epsilon} \delta^{(4)}(\vec{p}_1 + \vec{k}_1 - \vec{p}) \delta^{(4)}(\vec{p} - \vec{p}_2 - \vec{k}_2) \quad (43)$$

$$\bar{\phi} = ig^2 \int d^4p \frac{\bar{v}_1\gamma_5 u_1 \bar{u}_2\gamma_5 v_2}{\vec{p}^2 - M^2 + i\epsilon} \delta^{(4)}(\vec{P} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{K}) \quad (44)$$

$$\bar{\phi} = i\delta^{(4)}(\vec{P} - \vec{K})g^2 \frac{\bar{v}_1\gamma_5 u_1 \bar{u}_2\gamma_5 v_2}{\vec{P}^2 - M^2} \quad (45)$$

FIG. 8. $\mathfrak{X}$ FIG. 9. $\mathfrak{Y}$.

$$\mathfrak{X} = i\delta^{(4)}(\vec{P} - \vec{K})g^2 \frac{\bar{v}_1\gamma_5 u_1 \bar{u}_2\gamma_5 v_2}{s - M^2} \quad (46)$$

$$\mathfrak{Y} = ig^2 \int d^4p \frac{\bar{u}_2\gamma_5 u_1 \bar{v}_1\gamma_5 v_2}{\vec{p}^2 - M^2 + i\epsilon} \delta^{(4)}(\vec{p}_1 - \vec{p}_2 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{k}_1 - \vec{k}_2) \quad (47)$$

$$\mathfrak{Y} = i\delta^{(4)}(\vec{P} - \vec{K})g^2 \frac{\bar{u}_2\gamma_5 u_1 \bar{v}_1\gamma_5 v_2}{(\vec{p}_1 - \vec{p}_2)^2 - M^2} \quad (48)$$

$$\mathfrak{Y} = i\delta^{(4)}(\vec{P} - \vec{K})g^2 \frac{\bar{u}_2\gamma_5 u_1 \bar{v}_1\gamma_5 v_2}{t - M^2} \quad (49)$$

$$\frac{\mathcal{M}}{g^2} = \frac{\bar{v}_1\gamma_5 u_1 \bar{u}_2\gamma_5 v_2}{s - M^2} - \frac{\bar{u}_2\gamma_5 u_1 \bar{v}_1\gamma_5 v_2}{t - M^2} \quad (50)$$

$$\boxed{\frac{\mathcal{M}}{g^2} = \bar{v}_1\gamma_5 \left(\frac{u_1 \bar{u}_2\gamma_5 v_2}{s - M^2} - \frac{v_2 \bar{u}_2\gamma_5 u_1}{t - M^2} \right)} \quad (51)$$

2. 2

I relabel $\gamma' \rightarrow G$. I use both my QED and Real Scalar Field notes for this problem. My notation comes from these notes which match Schwartz. The interaction Lagrangian is

$$\mathcal{L}_1 = g\bar{\psi}\psi\phi + iG\bar{\psi}\gamma_5\psi\pi. \quad (52)$$

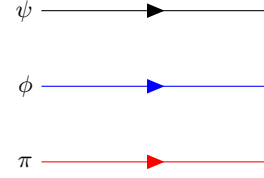
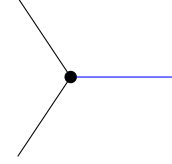


FIG. 10. Lines for different fields.

FIG. 11. Vertex for $g\bar{\psi}\psi\phi$. Each instance contributes the factor ig .

2.1. Feynman Rules

Fig. 10 assigns lines to fields. The propagators are

$$\begin{aligned} \psi : & \quad i \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon}, \\ \phi : & \quad \frac{i}{\vec{p}^2 - M^2 + i\epsilon}, \\ \pi : & \quad \frac{i}{\vec{p}^2 - \mu^2 + i\epsilon}. \end{aligned} \quad (53)$$

The two terms in $\mathcal{L}_1$ yield the two vertices in Figs. 11 & 12.

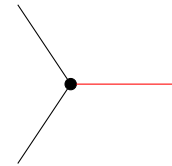
2.2. (1) $e^- \phi \rightarrow e^- \pi$

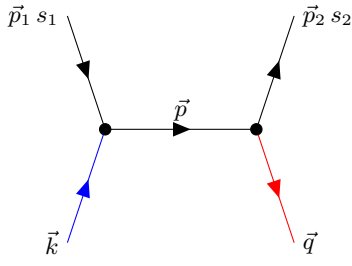
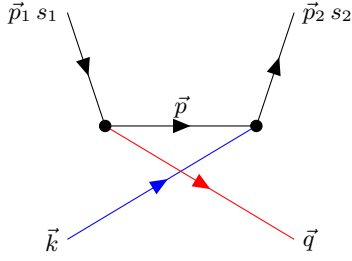
$$\vec{p}_1(e^-) + \vec{k}(\phi) \rightarrow \vec{p}_2(e^-) + \vec{q}(\pi) \quad (54)$$

$$\begin{aligned} s &= (\vec{p}_1 + \vec{k})^2 = (\vec{p}_2 + \vec{q})^2, \\ t &= (\vec{p}_1 - \vec{p}_2)^2 = (\vec{k} - \vec{q})^2, \\ u &= (\vec{p}_1 - \vec{q})^2 = (\vec{k} - \vec{p}_2)^2. \end{aligned} \quad (55)$$

$$\vec{P} = \vec{p}_1 + \vec{k}, \quad \vec{K} = \vec{p}_2 + \vec{q} \quad (56)$$

$$i\delta^{(4)}(\vec{P} - \vec{K})\mathcal{M} = \mathfrak{X} + \mathfrak{Y} \quad (57)$$

FIG. 12. Vertex for $iG\bar{\psi}\gamma_5\psi\pi$. Its factor is $-G\gamma_5$.

FIG. 13. $\mathfrak{Y}$ FIG. 14. $\mathfrak{Y}$

Feynman diagrams in Figs. 13 & 14. I use the shorthand $u_{s_n}(\vec{p}_n) = u_n$.

$$\mathfrak{Y} = gG \int d^4p \bar{u}_2 \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} u_1 \delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{p}_2 - \vec{q}) \quad (58)$$

How did I decide whether γ_5 goes to the left or right of the propagator? It should go on the same side as the external electron, $\bar{u}_2$, in its vertex.

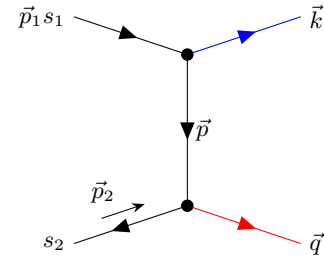
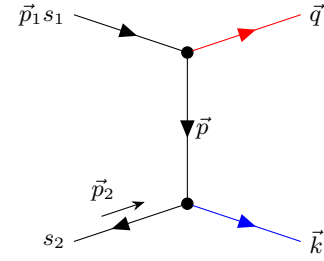
$$\mathfrak{Y} = gG \int d^4p \bar{u}_2 \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} u_1 \delta^{(4)}(\vec{P} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{K}) \quad (59)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) gG \bar{u}_2 \gamma_5 \frac{\not{P} + m}{s - m^2} u_1 \quad (60)$$

$$\mathfrak{Y} = gG \int d^4p \bar{u}_2 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} \gamma_5 u_1 \delta^{(4)}(\vec{p}_1 - \vec{q} - \vec{p}) \delta^{(4)}(\vec{p} + \vec{k} - \vec{p}_2) \quad (61)$$

$$\mathfrak{Y} = gG \bar{u}_2 \frac{\not{p}_1 - \not{q} + m}{(\vec{p}_1 - \vec{q})^2 - m^2} \gamma_5 u_1 \delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{q} - \vec{p}_2) \quad (62)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) gG \bar{u}_2 \frac{\not{p}_1 - \not{q} + m}{u - m^2} \gamma_5 u_1 \quad (63)$$

FIG. 15. $\mathfrak{Y}$.FIG. 16. $\mathfrak{Y}$.

$$i\mathcal{M} = gG \bar{u}_2 \gamma_5 \frac{\not{p} + m}{s - m^2} u_1 + gG \bar{u}_2 \frac{\not{p}_1 - \not{q} + m}{u - m^2} \gamma_5 u_1 \quad (64)$$

Because only $|\mathcal{M}|^2$ is observable I drop the overall factor of i .

$$\mathcal{M} = gG \bar{u}_2 \left(\gamma_5 \frac{\not{p} + m}{s - m^2} + \frac{\not{p}_1 - \not{q} + m}{u - m^2} \gamma_5 \right) u_1 \quad (65)$$

2.3. (2) $e^- e^+ \rightarrow \pi \phi$

$$\vec{p}_1(e^-) + \vec{p}_2(e^+) \rightarrow \vec{q}(\pi) + \vec{k}(\phi) \quad (66)$$

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{K} = \vec{k} + \vec{q} \quad (67)$$

$$\begin{aligned} s &= (\vec{p}_1 + \vec{p}_2)^2 = (\vec{q} + \vec{k})^2, \\ t &= (\vec{p}_1 - \vec{q})^2 = (\vec{p}_2 - \vec{k})^2, \\ u &= (\vec{p}_1 - \vec{k})^2 = (\vec{p}_2 - \vec{q})^2. \end{aligned} \quad (68)$$

$$i\delta^{(4)}(\vec{P} - \vec{K}) \mathcal{M} = \mathfrak{Y} + \mathfrak{Y} \quad (69)$$

Feynman diagrams in Figs. 15 & 16.

$$u_{s_n}(\vec{p}_n) = u_n, \quad v_{s_n}(\vec{p}_n) = v_n \quad (70)$$

$$\begin{aligned} \mathfrak{Y} = gG \int d^4p \bar{v}_2 \gamma_5 \frac{\not{p} + m}{p^2 - m^2 + i\epsilon} u_1 \\ \delta^{(4)}(\vec{p}_1 - \vec{k} - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{q}) \end{aligned} \quad (71)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) gG \bar{v}_2 \gamma_5 \frac{\not{p}_1 - \not{k} + m}{(\vec{p}_1 - \vec{k})^2 - m^2} u_1 \quad (72)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) gG \bar{v}_2 \gamma_5 \frac{\not{p}_1 - \not{k} + m}{u - m^2} u_1 \quad (73)$$

$$\begin{aligned} \mathfrak{Y} = gG \int d^4p \bar{v}_2 \frac{\not{p} + m}{p^2 - m^2 + i\epsilon} \gamma_5 u_1 \\ \delta^{(4)}(\vec{p}_1 - \vec{q} - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{k}) \end{aligned} \quad (74)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) gG \bar{v}_2 \frac{\not{p}_1 - \not{q} + m}{(\vec{p}_1 - \vec{q})^2 - m^2} \gamma_5 u_1 \quad (75)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) gG \bar{v}_2 \frac{\not{p}_1 - \not{q} + m}{t - m^2} \gamma_5 u_1 \quad (76)$$

$$i\mathcal{M} = gG \bar{v}_2 \gamma_5 \frac{\not{p}_1 - \not{k} + m}{u - m^2} u_1 + gG \bar{v}_2 \frac{\not{p}_1 - \not{q} + m}{t - m^2} \gamma_5 u_1 \quad (77)$$

I drop the i .

$$\boxed{\mathcal{M} = gG \bar{v}_2 \left(\gamma_5 \frac{\not{p}_1 - \not{k} + m}{u - m^2} + \frac{\not{p}_1 - \not{q} + m}{t - m^2} \gamma_5 \right) u_1} \quad (78)$$

2.4. (3) $\pi\phi \rightarrow e^-e^+$

$$\vec{k}(\phi) + \vec{q}(\pi) \rightarrow \vec{p}_1(e^-) + \vec{p}_2(e^+) \quad (79)$$

$$\vec{P} = \vec{k} + \vec{q}, \quad \vec{K} = \vec{p}_1 + \vec{p}_2 \quad (80)$$

$$\begin{aligned} s &= (\vec{k} + \vec{q})^2 = (\vec{p}_1 + \vec{p}_2)^2, \\ t &= (\vec{k} - \vec{p}_1)^2 = (\vec{q} - \vec{p}_2)^2, \\ u &= (\vec{k} - \vec{p}_2)^2 = (\vec{q} - \vec{p}_1)^2. \end{aligned} \quad (81)$$

$$i\delta^{(4)}(\vec{P} - \vec{K}) \mathcal{M} = \mathfrak{Y} + \mathfrak{Y} \quad (82)$$

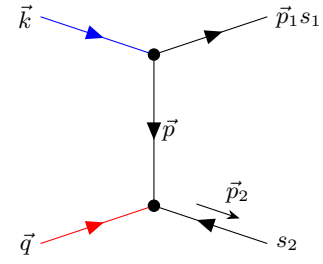


FIG. 17. $\mathfrak{Y}$.

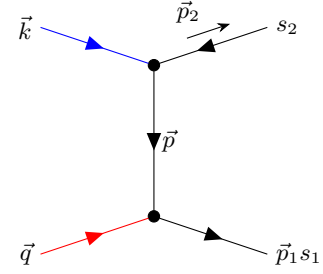


FIG. 18. $\mathfrak{Y}$.

Feynman diagrams in Figs. 17 & 18.

$$u_{s_n}(\vec{p}_n) = u_n, \quad v_{s_n}(\vec{p}_n) = v_n \quad (83)$$

$$\begin{aligned} \mathfrak{Y} = Gg \int d^4p \bar{u}_1 \frac{\not{p} + m}{p^2 - m^2 + i\epsilon} \gamma_5 v_2 \\ \delta^{(4)}(\vec{k} - \vec{p}_1 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{q} - \vec{p}_2) \end{aligned} \quad (84)$$

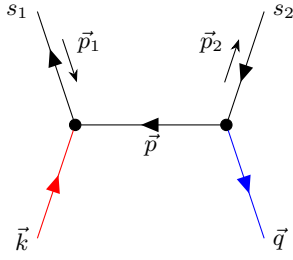
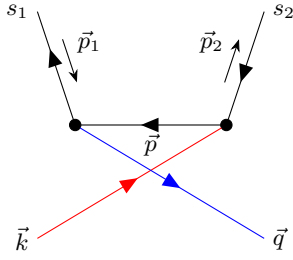
$$\mathfrak{Y} = Gg \bar{u}_1 \frac{\not{k} - \not{p}_1 + m}{(\vec{k} - \vec{p}_1)^2 - m^2} \gamma_5 v_2 \delta^{(4)}(\vec{P} - \vec{K}) \quad (85)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) Gg \bar{u}_1 \frac{\not{k} - \not{p}_1 + m}{t - m^2} \gamma_5 v_2 \quad (86)$$

$$\begin{aligned} \mathfrak{Y} = Gg \int d^4p \bar{u}_1 \gamma_5 \frac{\not{p} + m}{p^2 - m^2 + i\epsilon} v_2 \\ \delta^{(4)}(\vec{k} - \vec{p}_2 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{q} - \vec{p}_1) \end{aligned} \quad (87)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) Gg \bar{u}_1 \gamma_5 \frac{\not{k} - \not{p}_2 + m}{(\vec{k} - \vec{p}_2)^2 - m^2} v_2 \quad (88)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) Gg \bar{u}_1 \gamma_5 \frac{\not{k} - \not{p}_2 + m}{u - m^2} v_2 \quad (89)$$

FIG. 19. $\mathfrak{X}$ FIG. 20. $\mathfrak{Y}$

$$i\mathcal{M} = Gg\bar{u}_1 \frac{\not{k} - \not{p}_1 + m}{t - m^2} \gamma_5 v_2 + Gg\bar{u}_1 \gamma_5 \frac{\not{k} - \not{p}_2 + m}{u - m^2} v_2 \quad (90)$$

I drop the i .

$$\mathcal{M} = Gg\bar{u}_1 \left(\frac{\not{k} - \not{p}_1 + m}{t - m^2} \gamma_5 + \gamma_5 \frac{\not{k} - \not{p}_2 + m}{u - m^2} \right) v_2 \quad (91)$$

2.5. (4) $e^+\pi \rightarrow e^+\phi$

$$\vec{p}_1(e^+) + \vec{k}(\pi) \rightarrow \vec{p}_2(e^+) + \vec{q}(\phi) \quad (92)$$

$$\vec{P} = \vec{p}_1 + \vec{k}, \quad \vec{K} = \vec{p}_2 + \vec{q} \quad (93)$$

$$\begin{aligned} s &= (\vec{p}_1 + \vec{k})^2 = (\vec{p}_2 + \vec{q})^2, \\ t &= (\vec{p}_1 - \vec{p}_2)^2 = (\vec{k} - \vec{q})^2, \\ u &= (\vec{p}_1 - \vec{q})^2 = (\vec{k} - \vec{p}_2)^2. \end{aligned} \quad (94)$$

$$i\delta^{(4)}(\vec{P} - \vec{K})\mathcal{M} = \mathfrak{X} + \mathfrak{Y} \quad (95)$$

Feynman diagrams in Figs. 19 & 20.

$$v_{s_n}(\vec{p}_n) = v_n \quad (96)$$

$$\mathfrak{X} = Gg \int d^4p \bar{v}_1 \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} v_2 \delta^{(4)}(\vec{p}_1 + \vec{k} + \vec{p}) \delta^{(4)}(-\vec{p} - \vec{p}_2 - \vec{q}) \quad (97)$$

$$\mathfrak{X} = Gg \int d^4p \bar{v}_1 \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} v_2 \delta^{(4)}(\vec{p}_1 + \vec{k} + \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 + \vec{q}) \quad (98)$$

$$\mathfrak{X} = Gg \int d^4p \bar{v}_1 \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} v_2 \delta^{(4)}(\vec{P} + \vec{p}) \delta^{(4)}(\vec{p} + \vec{K}) \quad (99)$$

$$\mathfrak{X} = Gg\bar{v}_1 \gamma_5 \frac{m - \not{P}}{(-\vec{P})^2 - m^2} v_2 \delta^{(4)}(-\vec{P} + \vec{K}) \quad (100)$$

$$\mathfrak{X} = \delta^{(4)}(\vec{P} - \vec{K}) Gg\bar{v}_1 \gamma_5 \frac{m - \not{P}}{\vec{P}^2 - m^2} v_2 \quad (101)$$

$$\mathfrak{X} = \delta^{(4)}(\vec{P} - \vec{K}) Gg\bar{v}_1 \gamma_5 \frac{m - \not{P}}{s - m^2} v_2 \quad (102)$$

$$\mathfrak{Y} = Gg \int d^4p \bar{v}_1 \gamma_5 \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} v_2 \delta^{(4)}(\vec{p} + \vec{p}_1 - \vec{q}) \delta^{(4)}(\vec{k} - \vec{p}_2 - \vec{p}) \quad (103)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) Gg\bar{v}_1 \gamma_5 \frac{\not{k} - \not{p}_2 + m}{(\vec{k} - \vec{p}_2)^2 - m^2} v_2 \quad (104)$$

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) Gg\bar{v}_1 \gamma_5 \frac{\not{k} - \not{p}_2 + m}{u - m^2} v_2 \quad (105)$$

$$i\mathcal{M} = Gg \left(\bar{v}_1 \gamma_5 \frac{m - \not{P}}{s - m^2} v_2 + \bar{v}_1 \gamma_5 \frac{\not{k} - \not{p}_2 + m}{u - m^2} v_2 \right) \quad (106)$$

I drop the i .

$$\mathcal{M} = Gg\bar{v}_1 \gamma_5 \left(\frac{m - \not{P}}{s - m^2} + \frac{\not{k} - \not{p}_2 + m}{u - m^2} \right) v_2 \quad (107)$$